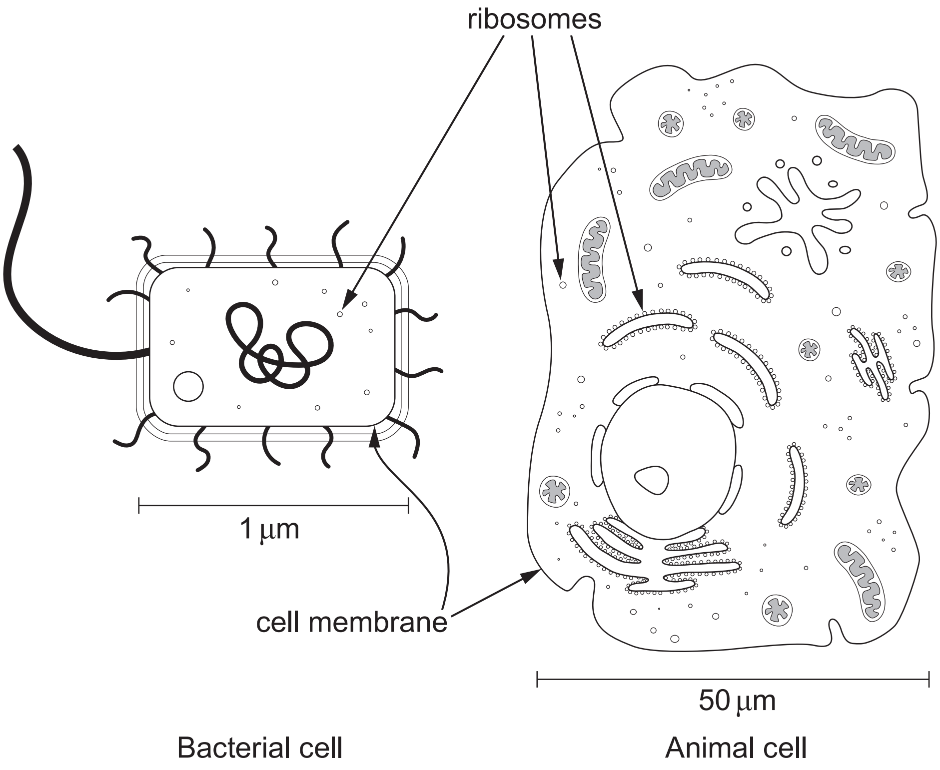
Answer **all** questions.

1. **Image 1.1** shows diagrams of two different cells. Scale bars are shown to illustrate the relative sizes of the cells.

Image 1.1



(a) Both cells produce the enzyme RNA polymerase. This is involved in the production of mRNA which is needed for translation by ribosomes.

(i) State the names of **two** other types of RNA involved in protein synthesis. [1]

....

(ii) Using your knowledge of cell structure, complete **Table 1.2** to compare the sites of translation and size of ribosomes in the two cells. [2]

Table 1.2

	Bacterial cell	Animal cell
location(s) of ribosomes involved in translation	....	....
size of ribosome	....	....

Examiner only

(b) Both cells contain membranes. The animal cell has considerably more membrane than the bacterial cell.
Use **Image 1.1** and your knowledge of the structure of these cells to suggest **two** reasons for this difference. [2]

....

....

....

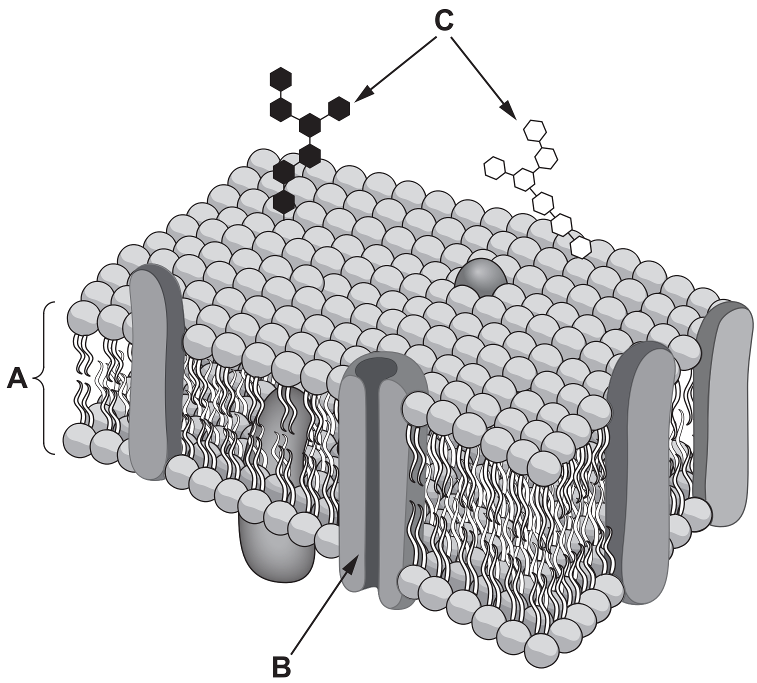
....

Question continued overleaf

Examiner only

(c) **Image 1.3** shows the cell surface membrane of the animal cell.

Image 1.3



(i) Identify components **A** and **B** in **Image 1.3 and** describe their functions. [3]

A ....

....

....

B ....

....

....

Examiner only

(ii) The components labelled **C** in **Image 1.3** form the glycocalyx and are involved in cell recognition. The cell surface membranes of the animal cell have a different composition from the inner mitochondrial membrane. The percentage composition of the three components in each membrane is shown in **Table 1.4**.

Table 1.4

Membrane	Percentage composition / %		
	Component **A**	Component **B**	Component **C**
animal cell surface membrane	53	42	5
inner mitochondrial membrane	22	78	0

Using the information in **Table 1.4** and your knowledge of the function of plasma membranes, suggest why there is:

I. no component **C** in the inner mitochondrial membrane; [1]

....

....

....

II. more component **B** in the inner mitochondrial membrane than the cell surface membrane. [1]

....

....

....

2400U101 05

10